

EXAM 7 – SPRING 2012

2. (3.5 points)

Given the following information as of December 31, 2011:

Accident Year	On-level Premiums	Cumulative Paid Loss	Fitted Paid Emergence Pattern
2008	\$1,300,000	\$600,000	70%
2009	1,200,000	350,000	45%
2010	1,200,000	200,000	25%
2011	1,300,000	75,000	10%

- Parameter standard deviation: 300,000
- Process variance/mean scale parameter (σ^2): 10,000

a. (2 points)

Estimate the total loss reserve using the Cape Cod method.

b. (0.5 point)

Calculate the process standard deviation of the reserve estimate in part a. above.

c. (1 point)

Calculate the total standard deviation and the coefficient of variation of the reserve estimate.

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Question 2 Sample Answer

Solution 1

a) OLEP × Paid Emergence

<u>AY</u>	<u>Used up EP</u>	<u>Paid Loss</u>
2008	910,000	600,000
2009	540,000	350,000
2010	300,000	200,000
2011	<u>130,000</u>	<u>75,000</u>
Total	1,880,000	1,225,000

$$\text{SB ELR} = 1.225\text{M}/1.880\text{M} = 65.159\% \quad (\text{Call this "x"})$$

$$\text{SB IBNR} = \sum (\text{ELR}) \times (1 - \% \text{ paid}) \times \text{OLEP}$$

$$\begin{aligned} &= .30 (1.3\text{M}) (x) + .55 (1.2\text{M}) (x) + .75 (1.2\text{M}) (x) + .90 (1.3\text{M}) (x) \\ &= 2,032,978.72 \end{aligned}$$

b) Process variance = $\sigma^2 \sum_i R_i = 10000(2032979)$

$$\text{Process SD} = \sqrt{(P \text{ Variance})} = 142,582.56$$

c) Total Variance = Process Variance + Parameter Variance
 $= 10000(2032979) + 300000^2$

$$\text{Total SD} = \sqrt{(\text{Total Variance})} = 332,159.28$$

$$\text{CV reserve estimate} = (\text{total SD})/(\sum_i R_i) = 332159.28/2032978.72 = .1634$$

Solution 2

a)

	(1)	(2)	(3)	(4)=(1)×(3)	(5)=(1)ELR(1-(3))
	<u>On lvl prem</u>	<u>paid loss</u>	<u>% paid</u>	<u>used up prem</u>	<u>loss reserve</u>
08	1,300,000	600,000	.70	910,000	254,122
09	1,200,000	350,000	.45	540,000	430,053
10	1,200,000	200,000	.25	300,000	586,436
11	1,300,000	<u>75,000</u>	.10	<u>130,000</u>	<u>762,367</u>
		1,225,000		1,880,000	2,032,978

$$\text{ELR} = \sum(2)/\sum(4) = 1,225,000/1,880,000 = .6516$$

$$\begin{aligned}
 \text{b) Process variance} &= \sigma^2 R \\
 &= 10,000(2,032,978) \\
 &= 2.033 \times 10^{10}
 \end{aligned}$$

$$\text{Process st. dev} = \sqrt{(2.033 \times 10^{10})} = 142,583$$

$$\begin{aligned}
 \text{c) Total st. dev} &= \sqrt{((2.033 \times 10^{10}) + 300,000^2)} = 332,159 \\
 \text{Coefficient of variation} &= \text{st. dev}/\text{mean} = 332,159/2,032,978 = .163
 \end{aligned}$$

Examiner Comment

The a. part of this question was a basic question on the Cape Cod methodology. Common errors included errors in calculation, and failure to recognize that the fitted emergence pattern was the % used (not the % unused).

The b. part requires the understanding that process variance = $\sigma^2 \times \text{mean}$ and that standard deviation is the square root of the variance. Common errors included misunderstanding of these concepts, as well as arithmetic errors.

The c. part requires the understanding that total variance is the sum of parameter and process variance, and that standard deviation is the square root of the variance. Common errors included adding the parameter standard deviation to the process deviation to find the total standard deviation, general misunderstanding of these concepts, as well as arithmetic errors.